



Module 12 Day 1: 10 Configuring firewall

Solution

The goal of this activity was to learn how to configure `firewalld`, a dynamically managed firewall that supports both network and firewall zones through trust-level definitions of network connections and interfaces.

Connect to the `firewalld` container with the command `docker exec -it firewalld bash`

1. List all available zones:
 - `sudo firewall-cmd --list-all-zones`
2. Set your `eth0` interface to your home zone:
 - `sudo firewall-cmd --zone=home --change-interface=eth0`
3. Display all active home zone services:
 - `sudo firewall-cmd --get-services`
4. Verify the home zone setting:
 - `sudo firewall-cmd --zone=home --list-all`
 - What is the status of IPV6, SSH, and Telnet?
 - IPV6: Allowed
 - SSH: Allowed
 - Telnet: Missing (not allowed)
 - Are we able to connect via SSH or Telnet?
 - We can connect via SSH, because it's allowed.
5. Log into the UFW container and connect to the `firewalld` IP:
 - `ssh 172.16.180.70 -l sysadmin`
 - password: cybersecurity
 - You should connect without issue.
 - Type `exit` to disconnect.
6. Test Telnet to `172.16.180.70`:

- Run `telnet 172.16.180.70`.
- What does the `no route to host` message mean?
 - This means that our `firewalld` program is blocking the connection.
- 7. Test pings to the `firewalld` IP.
ping `firewalld` IP
- 8. Switch back to the `firewalld` machine and block pings and ICMP requests in your home zone:
 - `sudo firewall-cmd --zone=home --add-icmp-block=echo-reply`
 - `sudo firewall-cmd --zone=home --add-icmp-block=echo-request`
- 9. List all the rules that are in place in your home zone:
 - `sudo firewall-cmd --zone=home --list-all`
 - Are all of our rules in place?
 - Yes.
- 10. Switch back to the `UFW` container and Run the command for ping that sends four packets to the `firewalld` machine.
 - `ping -c 4 172.16.180.70`
 - Are your ping requests blocked?
 - Yes.

Bonus:

- Using `rich-rules`, block the `UFW` server's IP address:
 - `sudo firewall-cmd --zone=home --add-rich-rule='rule family="ipv4" source address="172.16.180.72" reject'`
- Test if the `UFW` server's connection is blocked by testing your `rich-rule` by trying to SSH into `172.16.180.72`.
 - `ssh 172.16.180.72 -l sysadmin`
 - Why do you think you're not able to connect?
 - Because SSH or port 22 is being blocked.